FRB No-Tune 10 Watt PLL Transmitter Kit

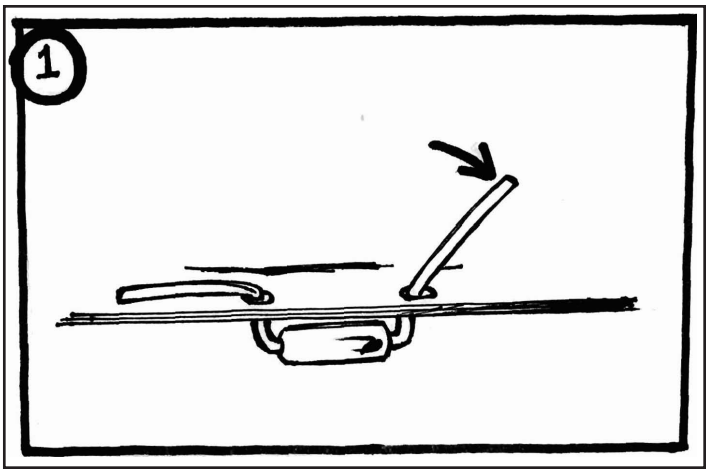
Tools required will be: Soldering iron 25 - 30 watts. Use a good qualtiy pencil style type with a plated tip not greater that 1/16" in size. Our preference is for Weller brand soldering irons. Do not use any soldering iron with a rating greater than 30 watts.

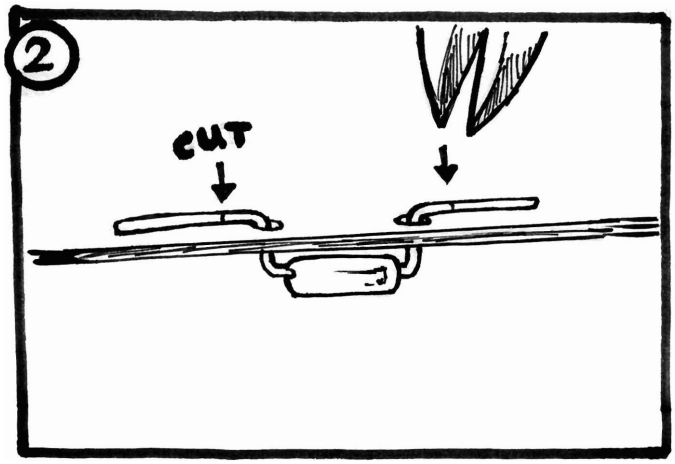
Electronic supply stores sell TV alignment tools which have a very small metal blade at the end of a plastic rod. This will be needed to adjust the fine frequency trimmer capacitor on the PLL. You will also need a small pair of diagonal cutters to trim the component leads after soldering. A pair of needle nose pliers will be very useful as well. It will be necessary to hold the circuit board steady while inserting the various components. A circuit board vise or "third hand" sold by Radio Shack and other stores will work quite well. If you don't want to buy one, you might want to cut a hole in a small cardboard box, just smaller than the board, so that the components leads can fit through, but the board is suspended. You can tape it down when soldering to keep it from moving.

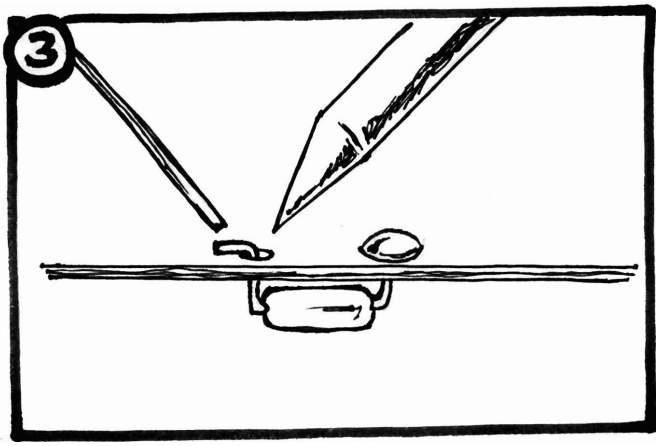
When soldering, heat the area surrounding the hole that the lead goes through, and the lead, from one side at an angle of about 45 degrees. After heating for a short time, apply the solder from the other side of the lead and hole. Move the solder in a circle and pull up the soldering iron with a wiping action when the solder melts. This gives you the best connection between the lead and the board. If you have trouble getting the solder to run, the temptation will be to melt the solder directly on the iron from one point. Instead, be sure to use thin solder. **Important.** In R.F. construction, solder components as close to the board as possible, unless otherwise instructed, or the circuit will be prone to instability due to extra lead length adding spurious inductance and capacitance..

Begin assembly by orienting the circuit board to match the loading diagram. Check to make sure that all the components are in the kit and match the parts list.

A component lead bender is required to properly form the resistor leads. These can be found at most electronic supply stores. Or, find a piece of plastic or something that can be cut to 4/10ths of an inch wide to form the leads of the resistors. Bend the leads 90 degrees to the body of the resistor.







For all leaded components, insert them into the circuit board as close to the board as possible. Bend the leads

90 degrees (parallel with the circuit board and resting on it) and cut the leads to no more than 1/8 inch long. This insures that the component will not fall back out when you turn it is time to solder. All components leads are treated in this manner unless noted otherwise. See diagram on page 1. When you bend a lead, bend it in the direction of the trace connected to the pad. Be careful not to bend the lead in a direction that allows it to touch a ground area

Take all the other resistors and insert them one by one into the circuit board, pulling the leads through so the resistor sits flush with the circuit board After all the resistors have been inserted into the circuit board at the proper places indicated, solder them into place after confirming proper placement according to the loading diagram.

Next go to the capacitors. Observe correct polarity on the electrolytic capacitors, they are polarized - one lead is negative and the other is positive. On the side of the capacitor will be either a plus (+) sign or a minus (-) to indicate polarity. That indicates the polarity of the lead directly underneath the marking, and the opposing lead will be either negative or positive depending on the marking. Orient and place according to the layout diagram. Note the color of the variable capacitor (trimmer). C51 is green. It sits next to C36-37. Install C47 with the flat side facing toward IC4. The small surface mount capacitors are already soldered to the board.

Now, to the inductors. Start with the wide band RF chokes (L1-L5). For L1-L4 insert with leads bent at 90 degrees like the resistors into the designated holes. L5 is mounted on the surface of the pads, no holes. Prepare in the same manner as L7. L6 has a small gray or beige or green body, a bit smaller than the resistors. Solder with one lead on the surface pad of the MAV11 and the other through the hole in the board at the juncture of R6 and C7. Keep L6 as close to the board as possible.

After you have inserted all of the resistors and capacitors with leads into the board. It is

L7 is a larger coil coated with a red enamel coating. It will be necessary to properly prepare the leads for soldering. Scrape the enamel off on the wire at the tip of each lead

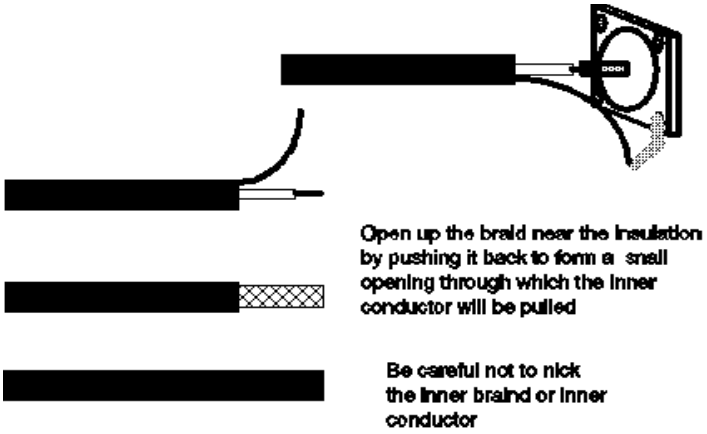
Grasp one lead of the component with needle nose pliers as close to the body of the component as possible. Then bend the wire lead at a right angle as shown above. Some components such as resistors and rf chokes will require two bends, as above. Always bend against the tip of the pliers. The internal connection can be compromised if pliers are not used as a strain relief when bending.

Installation of semiconductors - integrated circuits, transistors and diodes. Observe correct orientation of the ICs, the dot and/or notch should be facing in the direction indicated on the loading diagram. Install and insert the 18 pin socket for IC4, PIC16F628. Install and insert the 8 pin sockets for IC1 & IC5, LF351/TL071. Insert 12 volt regulator IC9, bend the leads 90 degrees down to match the hole pattern. Insert 5 volt regulators 78L05 IC3/5, observe correct orientation of the flat side, facing toward the edge of the board. Place IC6, MAV11, as indicated on the placement diagram, gently push each lead down and solder. Insert IC7, do not push all the way down, **leave about 1/16" between the POS150 and the surface of the board**. Observe correct orientaion, pin with blue dot goes to top left. Q1 will be last component to be attached to the circuit board - see the instructions whcih follow on the next page. Insert D1-D4, observe correct orientation. Position with black bands matching diagram. Solder all these components, diodes, transistors and ICs very carefully - do not apply heat for very long to each pin, just long enough to get a good shiny joint without excess solder.

Install the 8 position dip switch, orient with postion #1 at top and solder carefully.

The last component to install is the 16 MHz crystal. Solder very carefully and do not apply heat very long.

Pick out the coaxial cable from the wires provided, it will be marked in most cases with the legend RG174 and should be black in color. The coaxial cable consists of an outer insulator covering a wire braid which surrounds another insulated inner conductor. Strip the outer insulation back about one inch and pull the inner wire through the braided section by taking a small pointed tool to open up a hole in the braid near the outer insulation through which the inner conductor is pulled using needle nose pliers. Be gentle. do not cut the braid or crush or nick the inner conductor in this process Strip the inner conductor back about 1/4 inch or less. Twist it tight and tin the end. Twist the outer braid so that is tight and there are not any stray pieces sticking out. The braid should be bent outwards so it is about 90 degrees to the inner conductor. Tin the end of the braid. Solder



the inner conductor to the pad marked RF. Grab the braid with a pair of needle nose pliers and solder it to the ground area next to the RF out pad. It is important to hold the braid with the pliers to prevent the heat from traveling up the braid and melting the braid into the inner conductor. This would short the output transistor and cause immediate failure.

Take one of the red/white twisted pair hookup wires provided, strip and tin one end and solder the red lead to the spot marked 13.5 volts on the layout diagram. Solder the white wire to the ground point. Take another twisted pair wire, strip and tin the ends, solder one lead to the point marked audio in and the other to the point next to it marked ground. Note which color goes where.

Now, check your work carefully, look for any solder bridges between pins or leads. Use a voltmeter or DVM with a contintuity test, this will help quite a bit. Using the continuity function (the meter will beep if the leads are shorted together), put one probe on a voltage trace and the other probe on the ground. You should not hear a beep, if you do there is a short. Keeping the probe on the ground, touch all the voltage traces, signal traces, Q1 pads, etc. There should not be any shorts present. If you find a short, carefully inspect the PCB for any shorts between a trace or pad and ground. Solder can bridge between two areas. Also, components lead that are too long can form a bridge to the ground area. All the solder joints should be nice and shiny without excessive solder.

With all the components with the exception of Q1 attached and leads connected to the circuit board, attach the provided heat sink to the 7 X 7 aluminum plate. See the diagram provided. Take the aluminum heat spreader provided in the kit and apply a thin film of thermal compound to one side of it and place it in the cut-out area with compound side on the heat sink. Align the hole in the spreader with the hole in the heat sink. Next, using the aluminum spacers and 4-40 nuts and bolts provided, mount the circuit board on the panel so the cut-out for Q1 matches the opening on the panel. Q1 will lie on its back against the heat sink copper heat spreader that sits beneath the circuit board. Install Q1 after trimming its leads so they extend about 1/4 of an inch beyond the edge of the opening in the circuit board and applying thermal compound to its back side. Place Q1 in the opening and line up its mounting hole up with the corresponding holes in the heat spreader and heat sink. Make certain the transistor's lead are aligned properly on the circuit pads, straight and parallel with no overlap to either side. Take the long 4-40 bolt provided and the small copper strap. Make a small hole in the middle of the strap and insert the bolt through it. Then insert the bolt from the top of the transistor so it passes through both the heat spreader and heat sink. Secure it on the other side of the heat sink with a 4-40 nut and screw. Hold Q1 in position while tightening it down. Position the copper strap so it goes straight across the tab of the transistor and contacts the ground area on both sides. When a bit of the thermal compound spreads out from around Q1 it is sufficiently tight. Then solder its leads to the two pads and the ground area in the middle. Finally, solder the copper strap to the ground area on both sides.

Now, again, starting from the back of the box. Attach the SO. 239 to the front of the panel (the side opposite from the one the PLL is mounted to) with the short 4-40 bolts and hex nuts. Place the supplied ground lug under one of the nuts and position as shown in the diagram. Pull the coaxial cable to the SO 239 connector and cut the R.F. wire so that it lays neatly in the box and does not have excessive length or does not drape across the circuit board. Leave about an extra 1 1/2 inch in the cable for attachment to the SO239. Prepare the coaxial cable as described above. Strip the outer jacket off to a distance of about 1 inch and strip the insulation off the inner conductor back 1/4 inch. Solder the inner conductor of the coaxial cable to the solder cup terminal in the middle of the SO239 connector and the RF. ground shield to the little solder lug, which goes under one of the nuts holding the SO239 to the chassis in order to ground it.

Next, install the banana jacks on 7X7 panel. Take the two twisted pair leads that go to the voltage and ground pads on the board, shorten the lengths so they lay nicely on the panel without excessive length. Strip the ends about 1/4 " and solder the red voltage lead to the center terminal of the fuse holder. Solder the white wire to the black banana socket. Solder a red piece of hook-up wire from the other termimal of the fuse holder to one termimal of the switch. Solder another short piece

of hookup wire from the other termimal of the switch to the red banana socket. Be certain that the leads are connected properly, a reversal of leads will destroy the device.

Mount the variable resistor on the front of the box from the inside of the box with the shaft facing outwards. There is a metal tab on the body of the variable resistor, cut this off flush. Now install the 1/4" RCA. jack in the middle hole in the box's front. with the outer lead tab facing toward the variable resistor. Connect a .001uf (102) capacitor from the center terminal of the RCA connector to the ground terminal of the RCA connector. Thread the lead through the ground lug and on over to the left terminal of the 10K variable resistor. Connect a 10uf electrolytic capacitor from the center terminal of the RCA connector to the center terminal of the 10K variable resistor. Postive lead of the capacitor goes to the RCA connector solder the RCA end. Solder the two capacitor leads at the center terminal of the RCA connector, the RCAground lug lead and the capacitor lead at the center termimal of the 10K variable resistor.Take the white ground lead of the twisted wire pair from the audio input pads of the PLL transmitter, strip 1/4" and solder to the left terminal of the variable resistor along with the ground lead going to the RCA connector.. Solder the red audio input wire to the center terminal of the variable resistor after stripping 1/4" of insulation.

This should complete the hookup of the transmitter to the chassis.

Your transmitter is now ready for preliminary testing. Use a short jumper of coax cable either RG58 or RG8X with PL259 plugs at both ends and connect the SO239 connector from the transmitter to the SO239 input connector of your power meter. Take another 1-2 ft coax jumper as described above and connect the output connector of the power meter to the input connector of a dummy load (you can use our 15-20 watt dummy load kit). A frequency counter should be used as well, put the probe of the frequency counter near the SO239 connector to which the transmitter is connected.

The coax jumper cables, frequency counter, power/SWR meter and dummy load can be purchased at many radio, TV and amateur radio supply stores. Be sure the dummy load, frequency counter and power/SWR meter are specified for the VHF frequency range, not CB freqeuncy range.

Now that everything is properly set-up it is time to apply power and check the transmitter for proper operation. Use a power supply that put out a voltage of 13.8 volts. Connect the red banana socket to the positive terminal (usually red) of the power supply with a banana plug and connect the black banana socket to the negative or ground terminal (usually black) of the power supply. Make sure the leads are connected properly before applying power. **NEVER RUN THE TRANSMITTER WITHOUT A LOAD, DOING OTHERWISE WILL LIKELY DAMAGE OR DESTROY THE OUTPUT TRANSISTOR**.

Turn on the power supply and switch on the transmitter. You should see some sort of movement of the power meter which should be set to a range of 0 to 25 watts. Check Q1, if it becomes extremely hot, turn off power immediately. Do the same if any other components become hot. The LED lock indicator on the board should be glowing. Turn off the power supply. Look for shorts caused by solder bridges or improperly placed components if any components become hot. Output power level is adjusted with the 10K trimmer on the PLL PCB. Square blue component.

Install IC1, IC4 & IC5 into their respective sockets. Do not use excessive force and be careful that all the leads go into the sockets and none bend under. Observe correct orientation of the ICs, the dot and/or notch should be facing in the direction indicated on the loading diagram. Go to the frequency setting chart and select your operating frequency. Set the dip switch according to the chart by sliding or pushing the switches to either an off or on condition. The switch setting represent a binary number which corresponds to the frequency of choice. **"0"** represents a switch closure or **on** while **"1"** represents a switch open or **off**. Check your settings, it is easy to get them reversed.

Turn the transmitter on. If you are lucky the transmitter will lock to frequency - the LED should should glow briefly and then dim if this is the case. You should see a power reading which can be adjusted with the 10K trimmer. With the correct frequency displayed on the frequency counter adjust the green trimmer capacitor next to IC4 to bring the frequency as close as possible to having all zeros from the second digit to the right of the decimal point (assuming your counter is set to display Mhz) i.e. 88.1 should read 88.10000. It will only take minute turns of the trimmer to do this. According to FCC regs the frequency tolerance should be + / - 2000 cycles. Using the 10K trimmer, adust the PLL transmitter to the deired power level. If you intend to use the PLL transmitter as an exciter to drive a larger power amplifier, adjust the level to the wattage required by the the power amplifier. For example, the FRB 150 watt amplifier requires around 8 watts and the 350 watt

amplifier rewuires 3-4 watts.

All of the above set-up procedures assume that the transmitter came up without any problems. If you are careful about your work, this should be the case. If it does not , go back and check your work. Be sure that voltage is present at the two 13.5 volt points on the board. On the far right pin of IC4 you should see 12 volts. Be certain that all components are inserted correctly, especially Q1 , IC1. IC4 & IC5. It is really a rather simple circuit, an RF oscillator controlled by the PLL followed by a buffer and amplifier. Check the dipswitch settings. It is a common error to have two of the switch lines shorted together with a solder bridge.

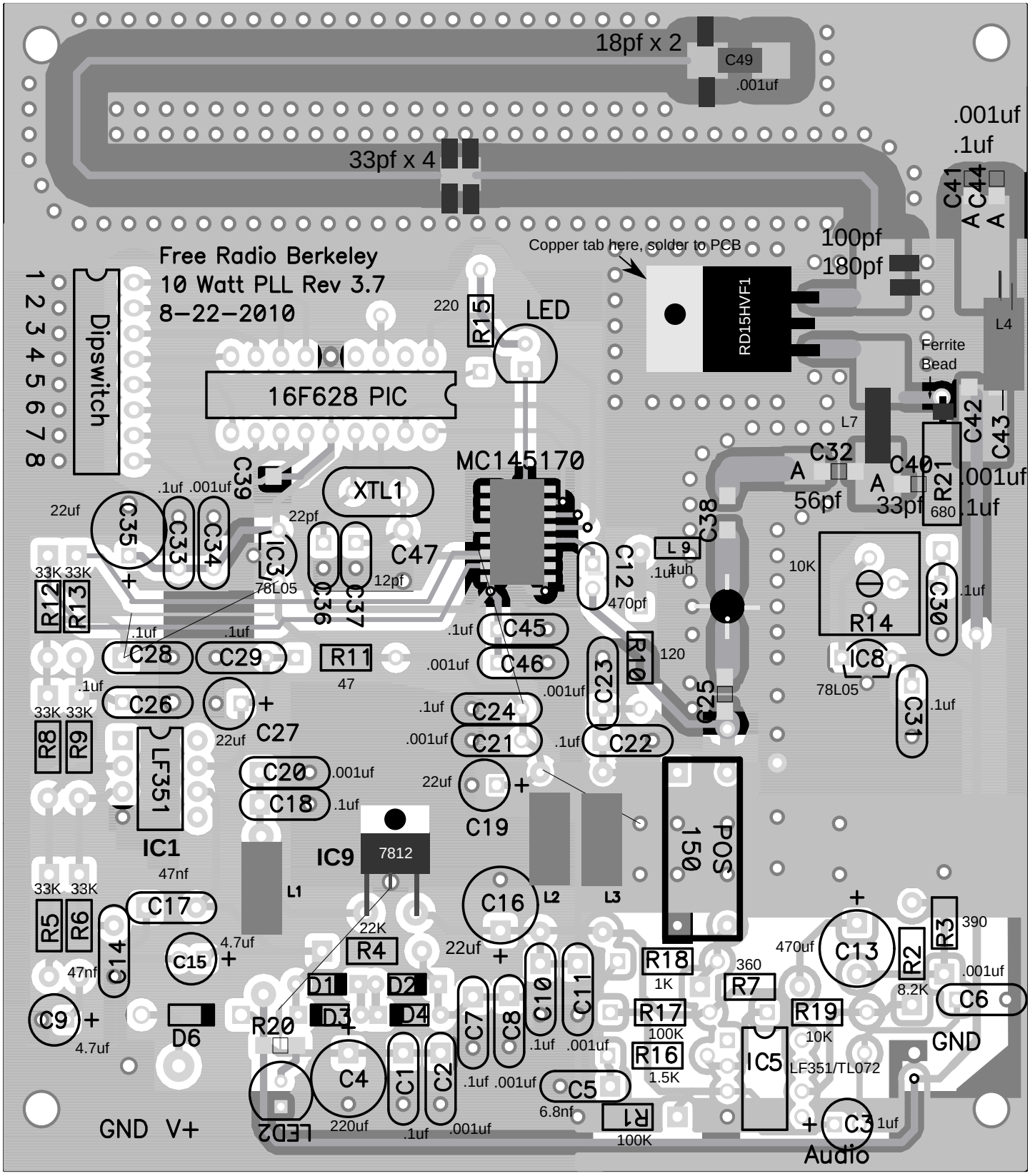
This should complete the hookup of the transmitter to the chassis. Repeat the tests (dummy load, frequency counter. etc.) above to check for proper operation. Open up the variable resistor about 3/4 of a turn, rotate all the way clockwise first and then turn back 1/4 or so. Tune an FM radio to the frequency chosen, a digital tuner works best. Turn the transmitter on, you should hear silence with very little hum if any. Run a line level audio source to the input of the transmitter, you should hear whatever your source material is.. If you do not hear anything, check your connections. Set the input level control at point where a line level signal input produces a signal on your receiver with a volume slightly less that of comparable stations. This will help prevent over modulating (a limiter is needed to fully prevent this) the transmitter.

It is very important to observe some basic broadcast engineering practices. Select a frequency that is not occupied by any other broadcaster. That frequency should have one channel of separation to either side in relation to other stations. In other words, if your are at 88.5, for example, 88.3 and 88.7 should be clear of broadcast stations. Use a good tuner and check the FM band from a number of different locations in the area. Check on both car and home tuners, use digital tuners. If you hear a weak signal, check the location, If you are outside the primary coverage area (30-50 miles) and have obstructions such as a hill between your area and the signal source, it will most likely be OK to use - double check though.

Next, use a filter. This is very important to prevent interference with other communications services. The first harmonic from an FM transmitter falls right into the VHF TV band (channels 6-12). Use the 7 element filter with transmitters 15 watts and under. Use the 9 element filter for those above that power. We have designs for other types of filters as well. The filter should be in a separate shielded metal enclosure, available as part of the filter kits from FRB. It is inserted in line with coax jumper cables between the transmitter and antenna. If you can find a friend with a spectrum analyzer, have them check the output of your transmitter for harmonics and such.

Over modulation can cause spurious emissions that interfere with adjacent channels. To prevent this use an audio limiter between the mixer and the transmitter. This can be purchased either from FRB or any well stocked pro-audio shop. Cost will range from $125 to $400.

By taking these measures we can deprive the FCC of their technical objections against micro power free radio. Do it right the first time and you will likely avoid scrutiny for sometime since the FCC usually becomes aware of a micro power operation from complaints.



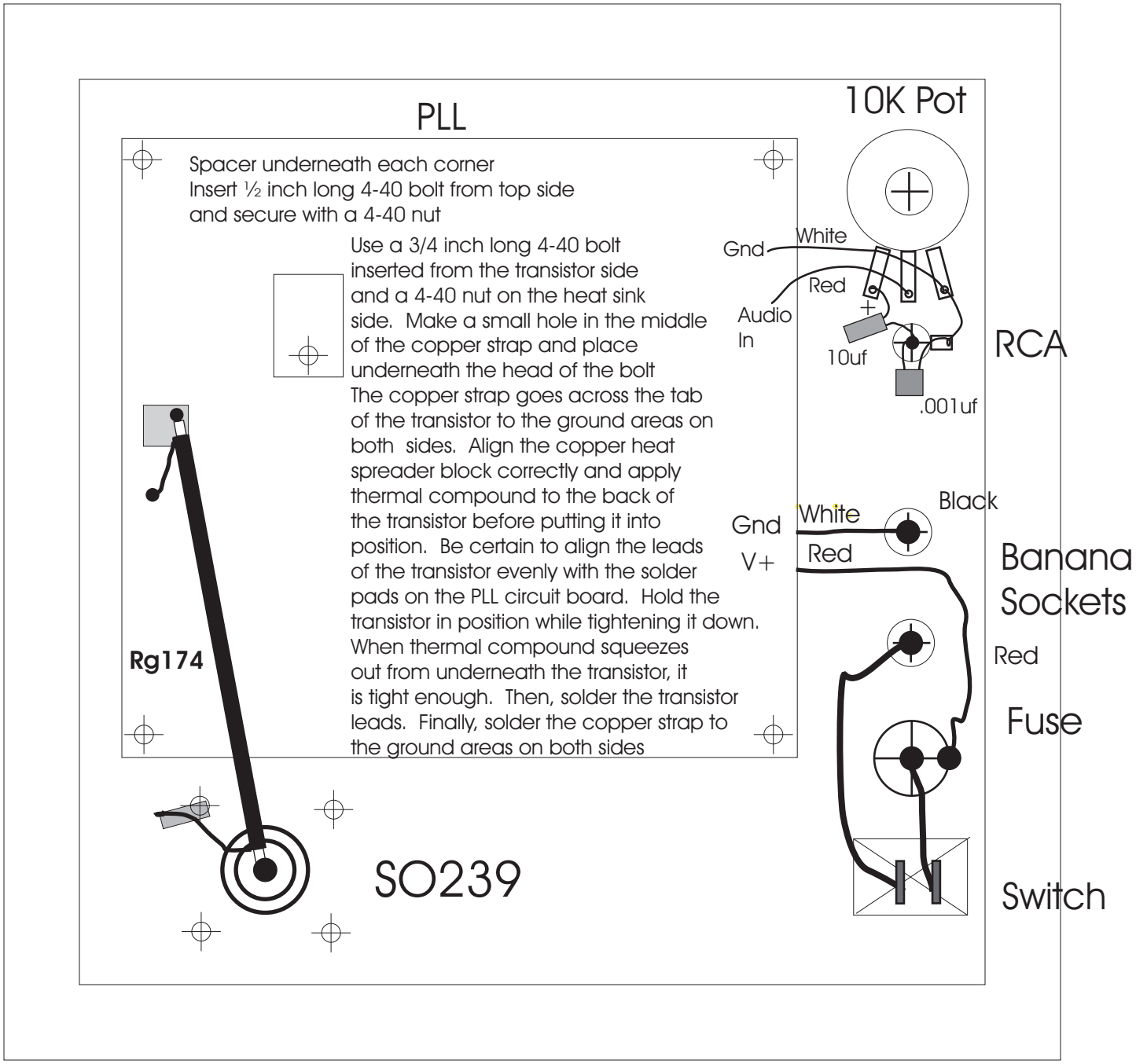
FRB 10 Watt PLL Component Placement

Serial 10 Watt PLL Rev 3.7 Parts List

Part number	Quantity	Description
C2, C6, C8, C11 C20-21, C23, C34, C46, C50,	10	.001uf capacitor, marked 102
C1, C7, C10, C18, C22, C24, C26, C28-29, C30-31, C45	12	.1uf capacitor, marked 104
C3	1	1uf electrolytic capacitor
C4	1	220uf electrolytic capacitor
C16, C19, C27, C35	3	22uf electrolytic capacitor
C9, C15	2	4.7uf electrolytic capacitor
C13	1	470uf electrolytic capacitor
C14, C17	2	.047uf capacitor, marked 473
C12	1	470pf capacitor, marked 471
C36	1	22pf capacitor, marked 22
C37	1	12pf capacitor, marked 12
C49	1	.001uf capacitor, larger
C47	1	5-30pf green trimmer capacitor
C52	1	10uf electrolytic capacitor
D1-D4	4	1N914 diode
D5	1	LED diode
IC1,5	2	LF351/TL071 op amp
IC2	1	MC145170 serial PLL
IC3, IC8	2	78L05 voltage regulator
IC4	1	PIC16F628A, programmed CPU
IC6	1	MAV11 amplifier
IC7	1	POS150 VCO, Mini-Circuits
IC9	1	7812 voltage regulator
L1-L4	4	Wide-band RF choke
L6	1	1uh choke, small brown or green body
L7	1	Coil 3 turns, #18, turns, .25 diameter
Q1	1	RD15HFV1 Transistor
R1-2, R17, R19	4	Resistor, 10K - brown, black, orange
R3	1	Resistor, 390 - orange, white, brown
R4	1	Resistor, 22K - red, red, orange
R5-6, R8-9, R12-13	6	Resistor, 33K - orange, orange, orange
R7	1	Resistor, 360 – orange, blue, brown
R10	1	Resistor, 120 - brown, red, brown
R11	1	Resistor 47 – yellow, violet black
R14	1	10K Trimmer
R15	1	Resistor, 220 - red, red, brown
R16, R18	2	Resistor, 1K – brown, black, red
R17	1	10K Potentiometer, audio level control
R21	1	Resistor, 680 - blue, gray, brown
X1	1	16 MHz crystal

Other items:
2 - 8 pin IC socket, 1 – 18 pin IC socket, 1 – 8 position dip switch – 1 ¾” #4 bolt - 4 – ¼” #4 bolt, 9 #4 nuts, 4 – ½” #4 bolts
4 #6 hex head sheet metal screws - 4 ¼” spacers, 1 - 10” length of #24 twisted pair, 1 – 5” length of #24 twisted pair, 1 – 10” length of RG174 coaxial cable, 1 – packet of thermal compound, 1 – red banana socket, 1 – red banana plug, 1 – black banana socket, 1 – black banana plug, 1 – SO239 socket, washer and nut for 10K potentiometer, 1 – RCA socket – heat sink – copper slug
spst power switch – transmitter mounting plate – fuse holder with fuse – 12” hook-up wire

10 Watt PLL Transmitter Wiring and Component Placement



Note: all connectors and components with the exception of the 10K potentiometer are mounted from the other side of the aluminum panel.

PLL Frequency Setting Chart

Frequency	DIPSwitch 87654321		Frequency	DIPSwitch 87654321	
87.9	49	00110001	92.8	98	01100010
88	50	00110010	92.9	99	01100011
88.1	51	00110011	93	100	01100100
88.2	52	00110100	93.1	101	01100101
88.3	53	00110101	93.2	102	01100110
88.4	54	00110110	93.3	103	01100111
88.5	55	00110111	93.4	104	01101000
88.6	56	00111000	93.5	105	01101001
88.7	57	00111001	93.6	106	01101010
88.8	58	00111010	93.7	107	01101011
88.9	59	00111011	93.8	108	01101100
89	60	00111100	93.9	109	01101101
89.1	61	00111101	94	110	01101110
89.2	62	00111110	94.1	111	01101111
89.3	63	00111111	94.2	112	01110000
89.4	64	01000000	94.3	113	01110001
89.5	65	01000001	94.4	114	01110010
89.6	66	01000010	94.5	115	01110011
89.7	67	01000011	94.6	116	01110100
89.8	68	01000100	94.7	117	01110101
89.9	69	01000101	94.8	118	01110110
90	70	01000110	94.9	119	01110111
90.1	71	01000111	95	120	01111000
90.2	72	01001000	95.1	121	01111001
90.3	73	01001001	95.2	122	01111010
90.4	74	01001010	95.3	123	01111011
90.5	75	01001011	95.4	124	01111100
90.6	76	01001100	95.5	125	01111101
90.7	77	01001101	95.6	126	01111110
90.8	78	01001110	95.7	127	01111111
90.9	79	01001111	95.8	128	10000000
91	80	01010000	95.9	129	10000001
91.1	81	01010001	96	130	10000010
91.2	82	01010010	96.1	131	10000011
91.3	83	01010011	96.2	132	10000100
91.4	84	01010100	96.3	133	10000101
91.5	85	01010101	96.4	134	10000110
91.6	86	01010110	96.5	135	10000111
91.7	87	01010111	96.6	136	10001000
91.8	88	01011000	96.7	137	10001001
91.9	89	01011001	96.8	138	10001010
92	90	01011010	96.9	139	10001011
92.1	91	01011011	97	140	10001100
92.2	92	01011100	97.1	141	10001101
92.3	93	01011101	97.2	142	10001110
92.4	94	01011110	97.3	143	10001111
92.5	95	01011111	97.4	144	10010000
92.6	96	01100000	97.5	145	10010001
92.7	97	01100001	97.6	146	10010010

The switch position is read right to left. “1” is off and “0” is on.

Frequency	DIPSwitch 87654321	
97.7	147	10010011
97.8	148	10010100
97.9	149	10010101
98	150	10010110
98.1	151	10010111
98.2	152	10011000
98.3	153	10011001
98.4	154	10011010
98.5	155	10011011
98.6	156	10011100
98.7	157	10011101
98.8	158	10011110
98.9	159	10011111
99	160	10100000
99.1	161	10100001
99.2	162	10100010
99.3	163	10100011
99.4	164	10100100
99.5	165	10100101
99.6	166	10100110
99.7	167	10100111
99.8	168	10101000
99.9	169	10101001
100	170	10101010
100.1	171	10101011
100.2	172	10101100
100.3	173	10101101
100.4	174	10101110
100.5	175	10101111
100.6	176	10110000
100.7	177	10110001
100.8	178	10110010
100.9	179	10110011
101	180	10110100
101.1	181	10110101
101.2	182	10110110
101.3	183	10110111
101.4	184	10111000
101.5	185	10111001
101.6	186	10111010
101.7	187	10111011
101.8	188	10111100
101.9	189	10111101
102	190	10111110
102.1	191	10111111
102.2	192	11000000
102.3	193	11000001
102.4	194	11000010
102.5	195	11000011

The switch position is read right to left. “1” is off and “0” is on.

Frequency	DIPSwitch 87654321	
102.6	196	11000100
102.7	197	11000101
102.8	198	11000110
102.9	199	11000111
103	200	11001000
103.1	201	11001001
103.2	202	11001010
103.3	203	11001011
103.4	204	11001100
103.5	205	11001101
103.6	206	11001110
103.7	207	11001111
103.8	208	11010000
103.9	209	11010001
104	210	11010010
104.1	211	11010011
104.2	212	11010100
104.3	213	11010101
104.4	214	11010110
104.5	215	11010111
104.6	216	11011000
104.7	217	11011001
104.8	218	11011010
104.9	219	11011011
105	220	11011100
105.1	221	11011101
105.2	222	11011110
105.3	223	11011111
105.4	224	11100000
105.5	225	11100001
105.6	226	11100010
105.7	227	11100011
105.8	228	11100100
105.9	229	11100101
106	230	11100110
106.1	231	11100111
106.2	232	11101000
106.3	233	11101001
106.4	234	11101010
106.5	235	11101011
106.6	236	11101100
106.7	237	11101101
106.8	238	11101110
106.9	239	11101111
107	240	11110000
107.1	241	11110001
107.2	242	11110010
107.3	243	11110011
107.4	244	11110100
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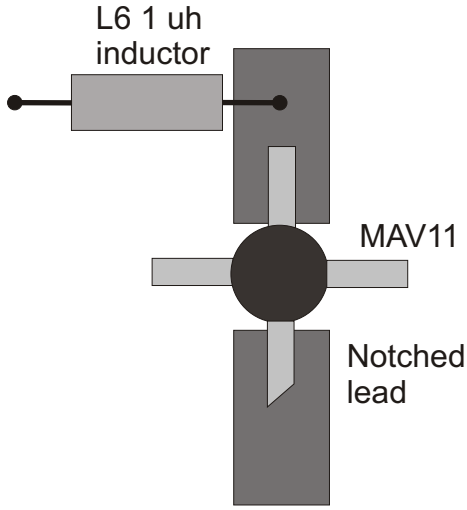
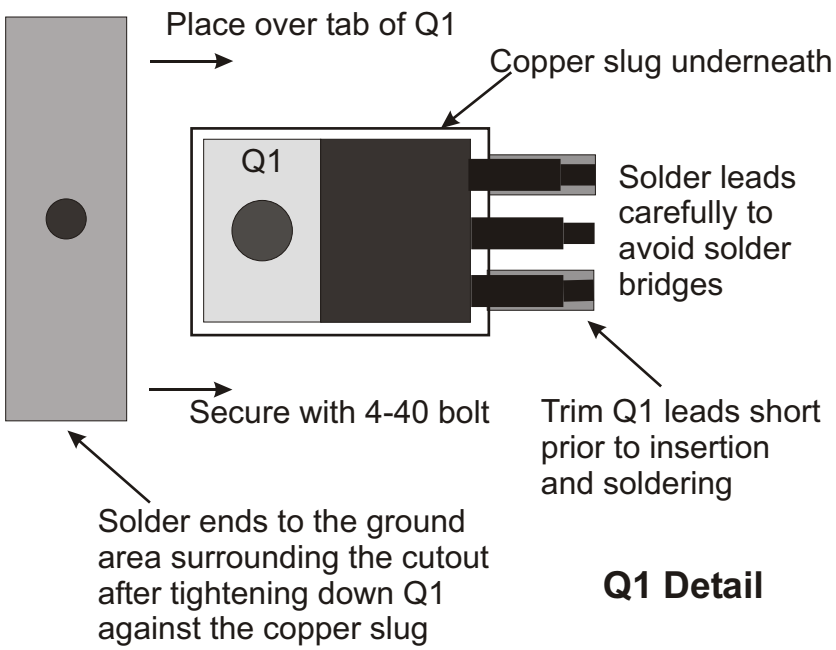
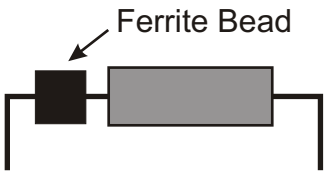
107.6	246	11110110
107.7	247	11110111
107.8	248	11111000
107.9	249	11111001



MAV11 and inductor detail



Q1 Detail



R16 Detail

10 Watt PLL Details